



PLOTDROP

SMART STORIES, ONE TITLE AWAY.



About Our Project

As fans of great stories, we're excited by the idea of reimagining plots — whether it's riffing on a familiar film or dreaming up something completely new.

Plotly gives scriptwriters a creative jumpstart by turning imaginative titles into baseline plots you can build from.





PlotDrop

Data Collection



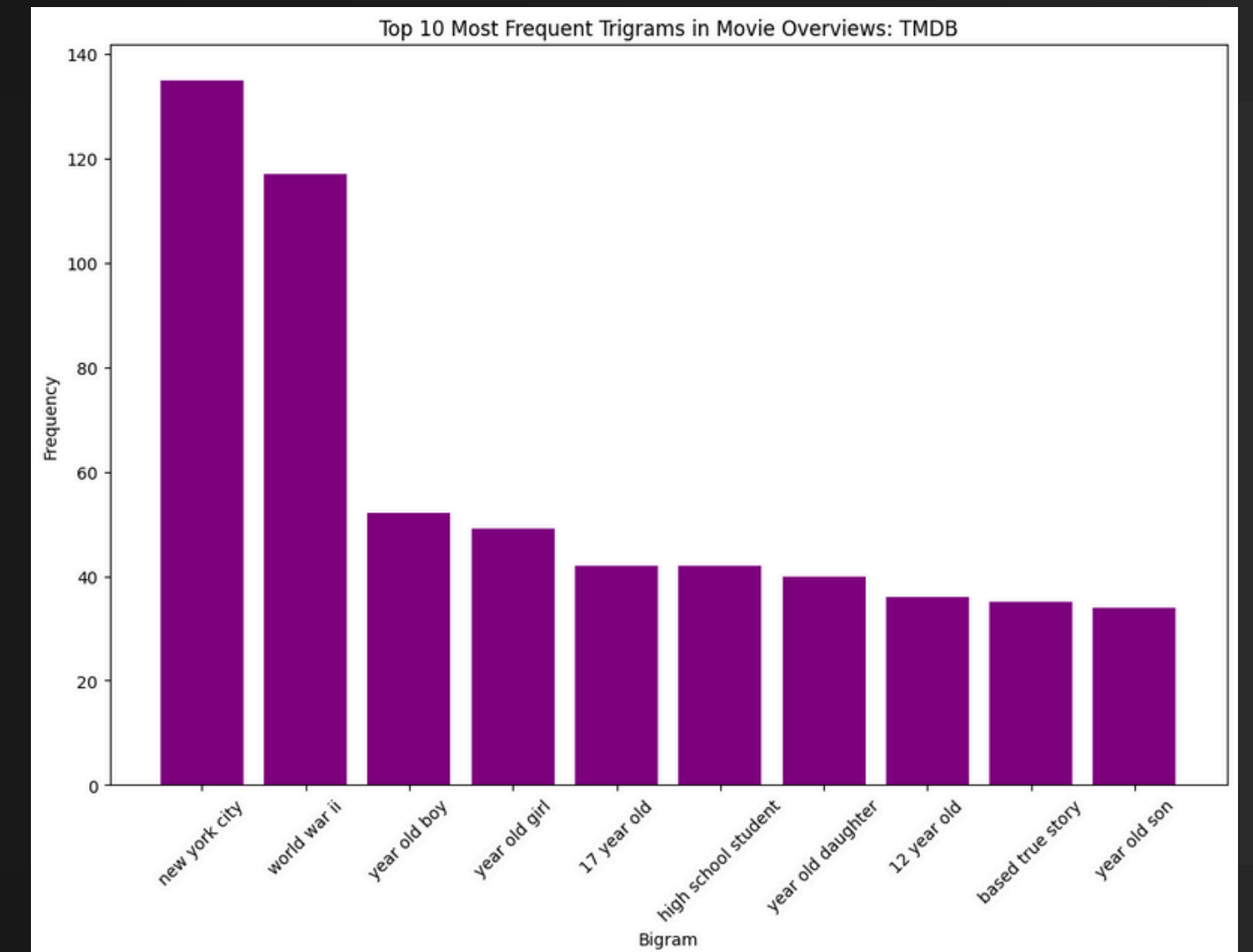
The Movie Data Base

Leveraged their API to pull the most popular and top rated movie plots from their database.
(42k rows)



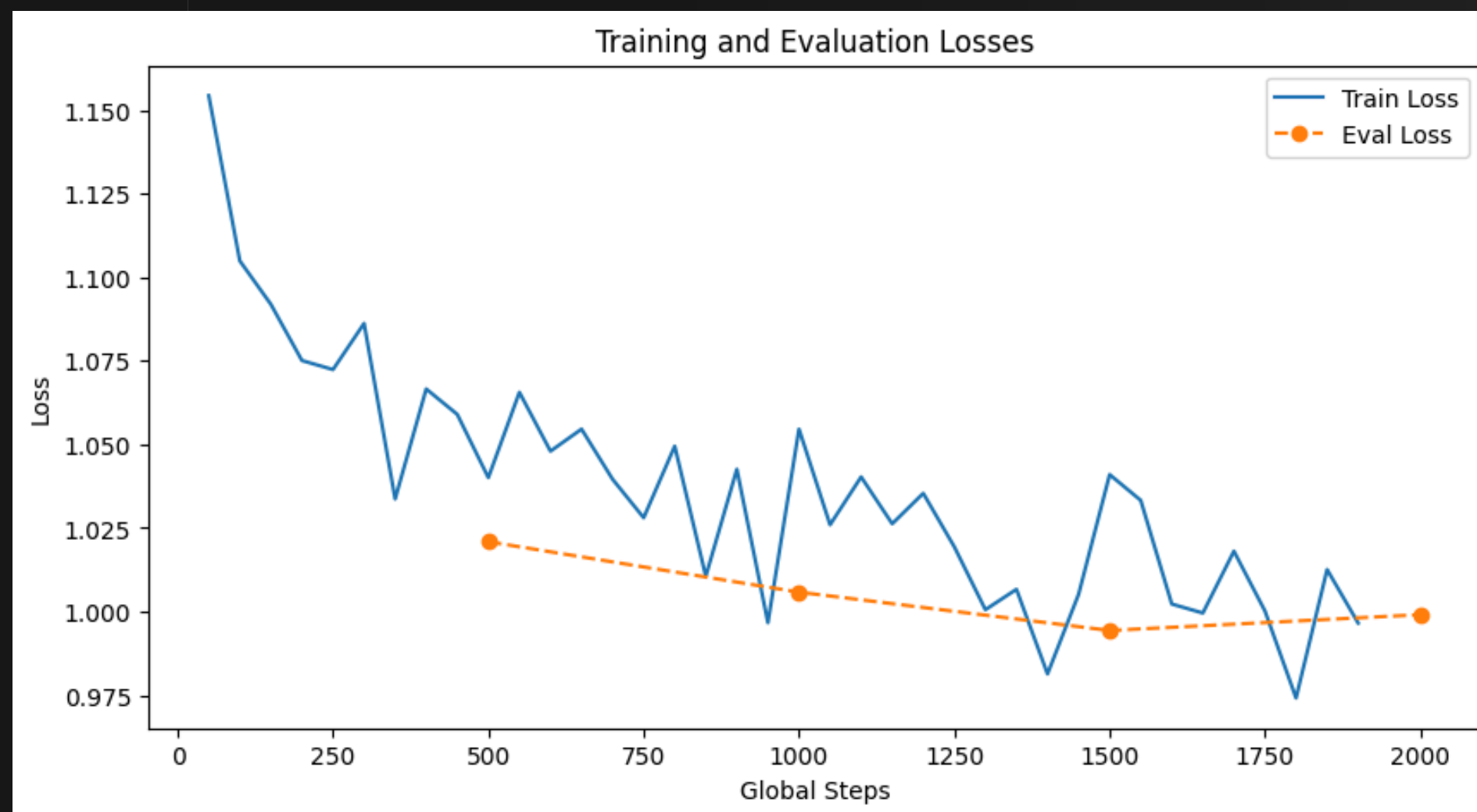
Wikipedia Movie Plots

Kaggle dataset download for movie titles, genres, and plots
(34k rows)





Model Development



Facebook / Bart Large - CNN

Input Example:

*Imagine a movie plot idea for **Forrest Gump***

Output Target:

A man with a low IQ has accomplished great things in his life and been present during significant historic events—in each case, far exceeding what anyone imagined he could do. But despite all he has achieved, his one true love eludes him.



The approach was to do a grid search for the best combination of key parameter values such as:

- Temperature
- Minimum Length
- Maximum Length
- Nucleus Sampling
- N-gram repeat

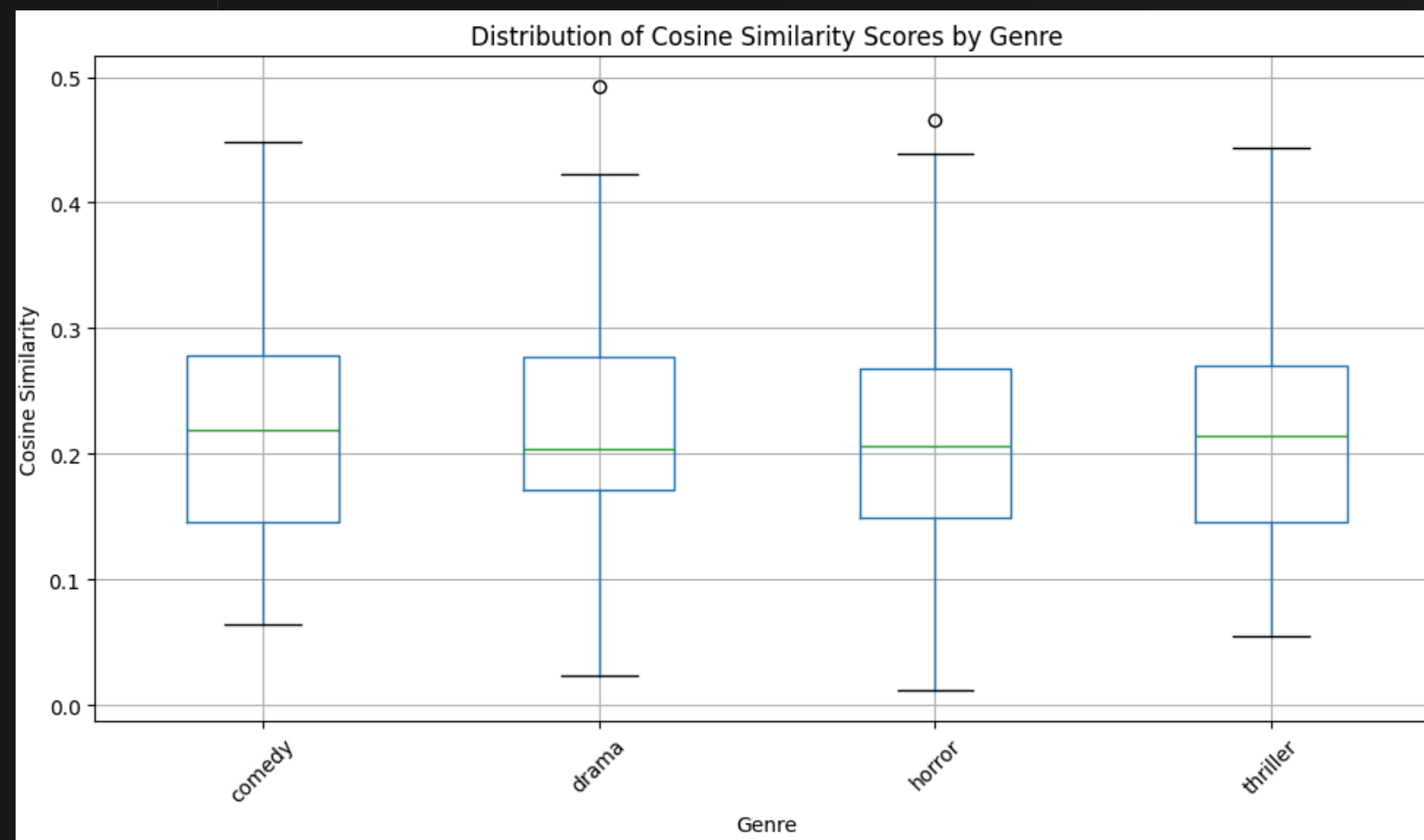
Top 5 configs by ROUGE-L F1:

	temperature	min_length	max_length	top_k	top_p	no_repeat_ngram_size	\
0	0.50	40	80	50	0.95		3
5	0.01	40	80	0	0.80		2
1	0.80	50	60	0	0.80		2
4	0.80	50	60	50	0.95		2
3	0.80	40	80	0	0.95		3

```
rougeL_f
0  0.164267
5  0.156882
1  0.144175
4  0.144095
3  0.138910
```



Baseline testing



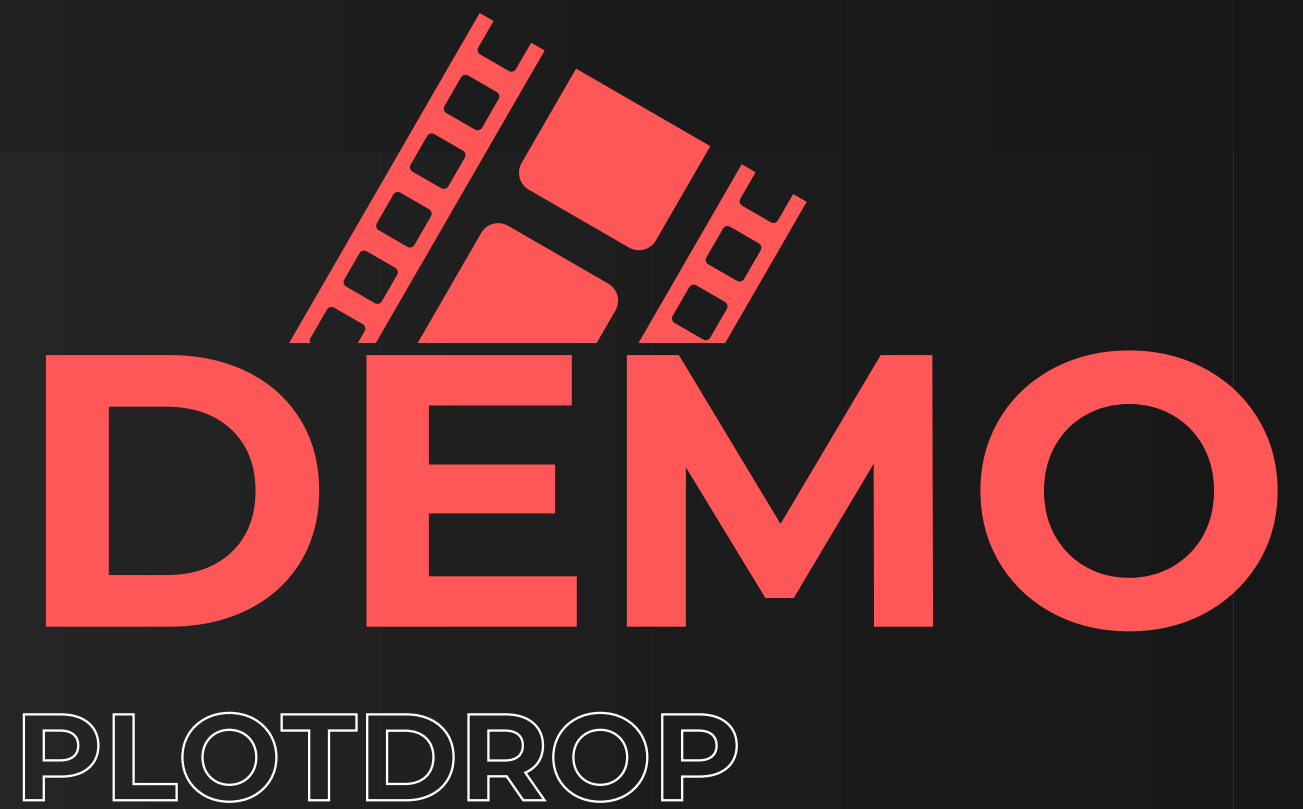
Cosine Similarity

We wanted to test creativity against the training data. Using the same titles, how similar were the generated outputs?

.49 Max Similarity

.21 Mean Similarity





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